STREAMLINED TIMELY MEDICINE SUPPLY TRACKER

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OUTLINE

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Abstract

- The Streamlined Timely Medicine Supply Tracker is a web-based medication management system designed to help users maintain consistent adherence to prescribed medicine schedules.
- The application combines an intuitive interface with real-time progress tracking, stock management, and automated reminders that alert users at scheduled medication times.
- A key feature is the medicine interaction warning system that analyses active medications and alerts users about potentially dangerous drug combinations.
- The system uses browser-based local storage for data persistence, ensuring privacy while maintaining access to medication schedules, emergency contacts, and allergy records.
- The responsive design incorporates HTML5, CSS3, and modern JavaScript for a seamless cross-device experience.

- The application addresses healthcare challenges through low-stock alerts, refill date tracking, and a snooze function for reminders.
- Advanced filtering and sorting allow users to organize medications by time, name, category, or status, while search functionality enables quick access to specific records.
- The interface includes accessibility features like ARIA labels, semantic HTML, and high-contrast emergency sections. Real-time weather integration provides context for temperature-sensitive medications, and print functionality allows users to generate physical medication schedules.
- By combining essential tracking with intelligent safety features in a privacy-focused design, the system helps users manage health regimens effectively while reducing medication errors and adverse interactions.

Existing System

- Medicine supply tracking was managed manually using registers, spreadsheets, or disconnected systems
- No real-time visibility into medicine stock levels, expiry dates, or supply chain status
- Frequent errors in stock management leading to medicine shortages or wastage due to expiry
- Lack of automated alerts for low stock or delayed supply deliveries
- Time-consuming and inefficient reporting process for administrators and pharmacists

Proposed System

- A web-based Medicine Supply Tracker providing real-time inventory management for healthcare facilities
- Interactive dashboard displaying live stock levels, expiry dates, and supply timelines
- Automated alerts for low stock levels, upcoming expiry, and delayed deliveries
- Streamlined data entry and reporting using HTML, CSS, and JavaScript for a responsive UI
- Role-based access for admins, pharmacists, and supply managers for organized workflow

System Architecture

FWD PROJECT:

- **Frontend (Client Side):** HTML5, CSS3, JavaScript for UI; browser renders the medicine tracker interface
- **Data Layer:** LocalStorage / JSON used to persist medicine inventory data on the client side

DSA PROJECT:

- **Application Layer:** Java-based backend logic implementing data structures for efficient data handling
- **Core Algorithms:** Sorting, searching, and queue-based scheduling algorithms implemented in Java

Medicine Tracker - System Architecture

USER INTERFACE

PRESENTATION LAYER (Frontend)

HTML Pages
- index.html (Home)
- tracker.html (Main)
- emergency.html
- reports.html

CSS Styling
- styles.css
- Responsive Design
- Flexbox/Grid Layouts
- Animations

Navigation
- Menu System
- Page Routing
- Breadcrumbs
- Mobile Toggle

UI Components
- Forms
- Tables
- Buttons
- Cards

LOGIC

BUSINESS LOGIC LAYER (JavaScript)

Medicine Manager
tracker.js
- addMedicine()
- markAsTaken()
- deleteMedicine()
- sortMedicines()
- checkInteractions()

Emergency Manager
emergency.html
- addContact()
- searchContact()
- removeContact()
- displayContacts()
- validateForm()

Reports Generator
reports.html
- displaySummary()
- calculateAdherence()
- showLowStock()
- exportToCSV()
- categoryReport()

Utility Functions
- formatTime() - Time formatting
- formatDate() - Date formatting
- validateForm() - Input validation
- setupReminders() - Notifications
- filterMedicines() - Search/Filter
- toggleMenu() - Navigation

DATA

DATA LAYER (Client-Side Storage)

Browser localStorage
Key-Value Storage
- medicines: JSON Array
- emergencyContact: String
- allergies: String
- lastReset: Date String

Persistent across sessions

In-Memory Storage
Runtime Variables
- medicines: Array[]
- snoozedMedicines: Object{}
- INTERACTION_WARNINGS
- Timer IDs for reminders

Lost on page refresh

Constants & Config
Static Data
- LOW_STOCK_THRESHOLD: 5
- SNOOZE_DURATION: 600000
- INTERACTION_WARNINGS: {}
- Date/Time formats

Immutable values

Project Modules

- **FWD - Medicine Inventory Module:** Add, update, and delete medicine records with name, quantity, batch number, and expiry date
- **FWD - Supply Tracking Module:** Monitor supply deliveries, track order status, and log supplier information
- **FWD - Alert & Notification Module:** Visual alerts for low stock and approaching expiry dates shown on the dashboard
- **DSA - Data Structure Module:** Java implementation of LinkedList, Queue, Stack, and Tree structures for data management
- **DSA - Algorithm Module:** Sorting (Merge Sort, Quick Sort) and searching (Binary Search) algorithms applied to medicine data

Objectives

- To design and develop a user-friendly web application for tracking medicine supply in real time
- To reduce medicine shortages and wastage through timely stock alerts and expiry monitoring
- To apply core data structures (LinkedList, Queue, Tree) and algorithms using Java for the DSA project
- To implement efficient sorting and searching operations on medicine inventory data
- To demonstrate integration of front-end web development skills with algorithmic problem-solving

COs Attainments - FWD

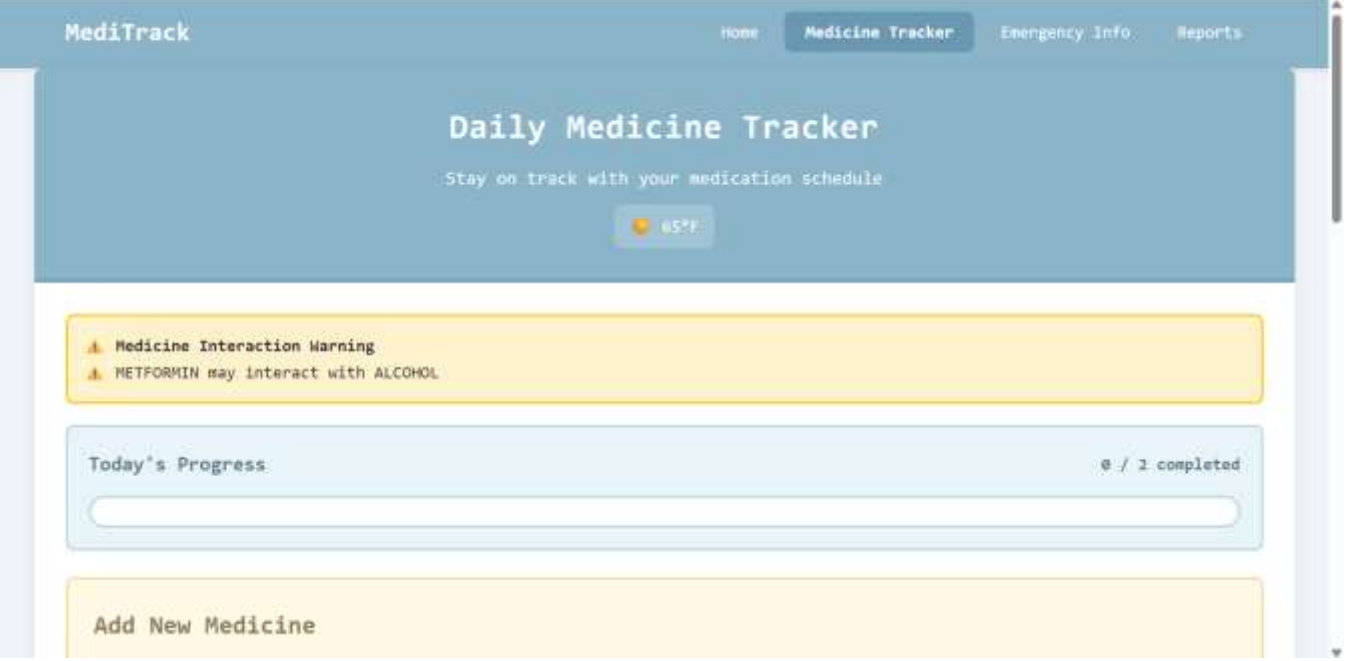
COURSE OUTCOMES	TOPICS
CO1	HTML Document Structure & Content, Headings, Paragraphs, Links; CSS syntax, Core selectors, Typography essentials; Color representations, Foundational spacing (padding, margin, border)
CO2	Forms & Form elements, Advanced input attributes, Input types; Forms validation, Semantic tags, Flexbox / CSS Grid, Media queries, CSS transitions / animations, Bootstrap CSS utilities or Responsive layouts without frameworks
CO3	Basic expressions, operators, Conditions, Loops; Functions, Arrow functions, Objects & Arrays, Array methods

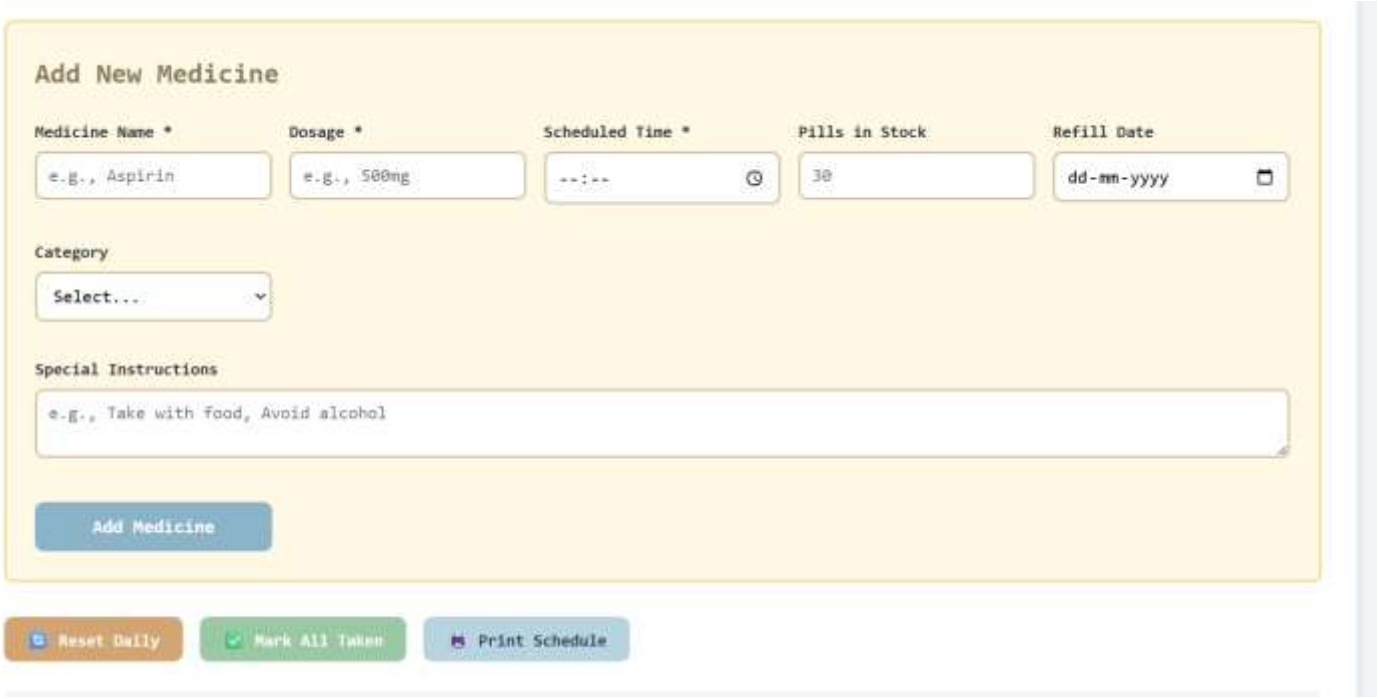
COURSE OUTCOMES	TOPICS
CO4	Event handling, DOM manipulation, Browser storage (LocalStorage), Asynchronous programming / Promises
CO5	Form validation with JavaScript, Handling user input dynamically, Optimize page load times, Implement meta tags for SEO-friendly pages, Deploying the Project

COs Attainments - DSA

COURSE OUTCOMES	TOPICS
CO1	Arrays, LinkedList, Stack, Queue used to manage medicine records or scheduling
CO2	Time and space complexity analysis (Big-O) applied to sorting/searching operations
CO3	Binary Search Tree used for ordered access to medicine/item data
CO4	Merge Sort / Quick Sort for sorting records; Binary Search for lookup
CO5	Complete Java project applying all DSA concepts to solve the problem

Screenshots for FWD & DSA


MediTrack
Home
Medicine Tracker
Emergency Info
Reports
Daily Medicine Tracker
Stay on track with your medication schedule
65°F
Medicine Interaction Warning
METFORMIN may interact with ALCOHOL
Today's Progress
0 / 2 completed
Add New Medicine


Add New Medicine
Medicine Name *
e.g., Aspirin
Dosage *
e.g., 500mg
Scheduled Time *
--:--
Pills in Stock
30
Refill Date
dd-mm-yyyy
Category
Select...
Special Instructions
e.g., Take with food, Avoid alcohol
Add Medicine
Reset Daily
Mark All Taken
Print Schedule

Search medicines...

Sort by Time

All Categories

Today's Medicines

Medicines to Take

Medicine	Dosage	Time	Stock	Notes	Status	Actions
No pending medicines						

Medicines Taken

Medicine	Dosage	Scheduled	Taken At	Notes	Actions
No medicines taken yet					

MediTrack

Home | Medicine Tracker | Emergency Info | Reports

🚨 Emergency Information

Critical information for emergencies

Emergency Contacts

Primary Emergency Contact *
Name

Contact Number *
(123) 456-7890

Relationship
Select...

Secondary Contact
Name

Secondary Number
(123) 456-7890

Primary Doctor
Dr. Smith

Doctor's Phone
(123) 456-7890

Preferred Hospital
City General Hospital

Insurance Information
Policy number, provider name, etc.

⚠️ Allergies & Medical Conditions

Drug Allergies
List all drug allergies (e.g., Penicillin, Aspirin)

Food Allergies
List food allergies (e.g., Peanuts, Shellfish)

Chronic Medical Conditions
Diabetes, Heart Disease, Asthma, etc.

Blood Type
Select...

Organ Donor Status
Select...

Additional Medical Notes

MediTrack
Home
Medicine Tracker
Emergency Info
Reports
Reports & Analytics
Track your medication adherence and patterns
0
Total Medicines
0%
Adherence Rate
0
Missed Doses
0
Low Stock Items
All Medicines
Medicine Name
Dosage
Schedule Time
Category
Current Stock
Refill Date
Status
No medicines tracked yet. Add your first medicine

No medicines tracked yet. Add your first medicine
Medicines by Category
No medicines to categorize yet.
Upcoming Refills
No refills needed in the next 30 days.
Print Report
Export to CSV

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        MEDICINE TRACKER SYSTEM - Java DSA

    Comprehensive Data Structures & Algorithms
            Application for Healthcare


╔═══════════════ MAIN MENU ═══════════════╗
╚═══════════════════════════════════════════╝

  1. Medicine Management
  2. Emergency Information
  3. Reports & Analytics
  4. Search & Sort Operations
  5. Advanced DSA Demonstrations
  6. Exit

Enter your choice: 1

╔═══════ MEDICINE MANAGEMENT ═══════╗
╚═══════════════════════════════════╝
  1. Add New Medicine
  2. View All Medicines
  3. Mark Medicine as Taken
  4. Delete Medicine
  5. Update Medicine Stock
  6. View Priority Queue
  7. Back to Main Menu

Choice: 1
```

```
Choice: 1

ADD NEW MEDICINE
═══════════════════════════════════════

Medicine Name: Aspirin
Dosage (e.g., 500mg): 500mg
Time (HH:MM format): 12:30
Stock quantity: 8
Category (optional):
Special instructions (optional):
Refill date (YYYY-MM-DD, optional):

Medicine added successfully!

╔═══════ MEDICINE MANAGEMENT ═══════╗
╚═══════════════════════════════════╝
  1. Add New Medicine
  2. View All Medicines
  3. Mark Medicine as Taken
  4. Delete Medicine
  5. Update Medicine Stock
  6. View Priority Queue
  7. Back to Main Menu

Choice: 3

Enter medicine name to mark as taken: Aspirin
Medicine marked as taken!

╔═══════ MEDICINE MANAGEMENT ═══════╗
╚═══════════════════════════════════╝
  1. Add New Medicine
  2. View All Medicines
  3. Mark Medicine as Taken
  4. Delete Medicine
  5. Update Medicine Stock
  6. View Priority Queue
  7. Back to Main Menu

Choice: ▌
```

Conclusion

- The FWD project successfully demonstrates a functional web-based medicine supply tracker with real-time inventory management
- The application effectively reduces manual effort, minimizes errors, and provides timely alerts for medicine stock management
- The DSA project reinforces understanding of core data structures and algorithm design through practical Java implementation
- Both projects together bridge front-end web development and algorithmic thinking, fulfilling the course outcomes effectively
- Future scope includes integrating a backend database and extending the DSA algorithms to handle larger datasets

Geo- Tag Photos

THANK YOU